Custom Encryption Data Guide

(Android Only)

Preface

Custom Encryption Data Guide
Version: V1.00.00
Status: [ ] Draft [★] Release [ ] Modify

Technical Support
If you encounter a problem while installing, registering or operating this product, please make sure that you have read the documentation. If you cannot resolve the Issue, please contact us.

Web: https://developer.pax.us

Release Notes

Date	Version	Author	Comment
2023-11-22	V1.00.00	Keenan.Wu	First Released.

1. Content

1. Custom Data Definition

1.1. Customize parameter filling rules

All custom data (that is, data provided by the customer that needs to be encrypted) need to be converted to HEX format.

The sensitive data that needs to be filled will be filled in a specific placeholder format, and the placeholder can use up to three of parameters to determine its function.

- The sensitive data placeholder starts with "${" and ends with "}".
- When the data format is incorrect, empty content or error message will be returned.

1.2. Parameter List

The parameters are divided into three levels. The first-level parameters are used to determine the data source, the second-level and third-level parameters are generally used to determine the format and conditions of the data.

1.2.1. First Parameters

PAN

The plain text PAN data.

Example: ${PAN}

PANLEN

The length of PAN data.

Example:

${PANLEN, HEX, 2} – replaced by the hex format length of PAN, and the length has been format to 2 digits.

${PANLEN, ASC, 1} - replaced by the asc format length of PAN, and the ASCII character corresponding to the length will be used as its value. If it exceeds 127, it will be returned as \xXX characters, usually it is 1-byte length.

${PANLEN} or ${PANLEN, DEC, 2} – replaced by the origin length of PAN, usually it is 2-digit decimal.

TRACK1DATA

The plain text track1 data.

Example: ${TRACK1DATA} - replaced by track1 data.

TRACK1LEN

The length of plain text track1, if Track1 does not exist, return 0.

Example:

${TRACK1LEN, HEX, 2} - replaced by the hex format length of track1, and the length has been format to 2 digits.

${TRACK1LEN, ASC, 1} - replaced by the asc format length of track1, and the ASCII character corresponding to the length will be used as its value. If it exceeds 127, it will be returned as \xXX characters, usually it is 1-byte length.

${TRACK1LEN} or ${TRACK1LEN, DEC, 2} – replaced by the origin length of track1, usually it is 2-digit decimal.

TRACK2DATA

The plain text track2 data.

Example: ${TRACK2DATA} - replaced by track2 data.

TRACK2LEN

The length of plain text track2, if Track2 does not exist, return 0.

Example:

${TRACK2LEN, HEX, 2} - replaced by the hex format length of track2, and the length has been format to 2 digits.

${TRACK2LEN, ASC, 1} - replaced by the asc format length of track2, and the ASCII character corresponding to the length will be used as its value. If it exceeds 127, it will be returned as \xXX characters, usually it is 1-byte length.

${TRACK2LEN} or ${TRACK2LEN, DEC, 2} – replaced by the origin length of track2, usually it is 2-digit decimal.

TRACK3DATA

The plain text track3 data.

Example: ${TRACK3DATA} - replaced by track3 data.

TRACK3LEN

The length of plain text track3, if Track3 does not exist, return 0.

Example:

${TRACK3LEN, HEX, 2} - replaced by the hex format length of track3, and the length has been format to 2 digits.

${TRACK3LEN, ASC, 1} - replaced by the asc format length of track3, and the ASCII character corresponding to the length will be used as its value. If it exceeds 127, it will be returned as \xXX characters, usually it is 1-byte length.

${TRACK3LEN} or ${TRACK3LEN, DEC, 2} – replaced by the origin length of track3, usually it is 2-digit decimal.

ENTRYMODE

Entry Mode, the length is 1 digit.

0 – For Manual.
1 – Swipe card.
2 – Contactless tap card.
3 – For Scan.
4 – Contact chip card.

Example: ${ENTRYMODE}

EXPIRYDATE

The expiry date for the card.

CVV

The CVV data for the card.

CVVLEN

The length of CVV data.

TAGLIST

To replace the plaintext tag list in Tag List field.

${TAGLIST} – if the Tag List field is "9F339F34", it replaced by "9F3303E0B8C89F34031F0200".

TLV

A complete TLV format, including TAG, LEN, and VALUE.

Example: ${TLV, 9F33} - replaced by "9F3303E0B8C8".

TLVVALUE

A TLV value only.

Example: ${TLVVALUE, 9F33} - replaced by "E0B8C8"。

KSN

The KSN for this transaction, it will be used in Mac Data Format.

1.2.2. Second parameters

HEX

Indicates that the value is filled in hexadecimal format (for LEN type).

Example: length=16, return “10”.

DEC

Indicates that the value is filled in decimal format (for LEN type).

Example: length=16, return “16”.

ASC

Indicates that the value is filled in ASCII format (for LEN type).

Example: length=16, return ‘\x10’.

1.2.3. Third parameters

Generally, it appears in the form of numbers, indicating a fixed number of data digits. If the actual value is less than the number of digits, 0 is added in front; it can also be used to indicate information such as index.

2. User Example

The customer needs to encrypt a string of data content as

012357 + length of Track2(HEX, 2 digits)+ Track2

The Custom Data format is

303132333537${TRACK2LEN, HEX, 2}${TRACK2DATA}

The encrypted format of this string of data will returned by BroadPOS P2PE:

01235723;5128570154494031=2311622000000000?

The customer needs to encrypt a string of data content as

0123\x5A + length of PAN with ASC, 1byte + PAN

The Custom Data format is

303132335A${PANLEN, ASC, 1}${PAN}

Or 30313233${TLV, 5A}

The encrypted format of this string of data will returned by BroadPOS P2PE (the data has been convert to HEX format):

“303132335A1035313238353730313534343934303331”